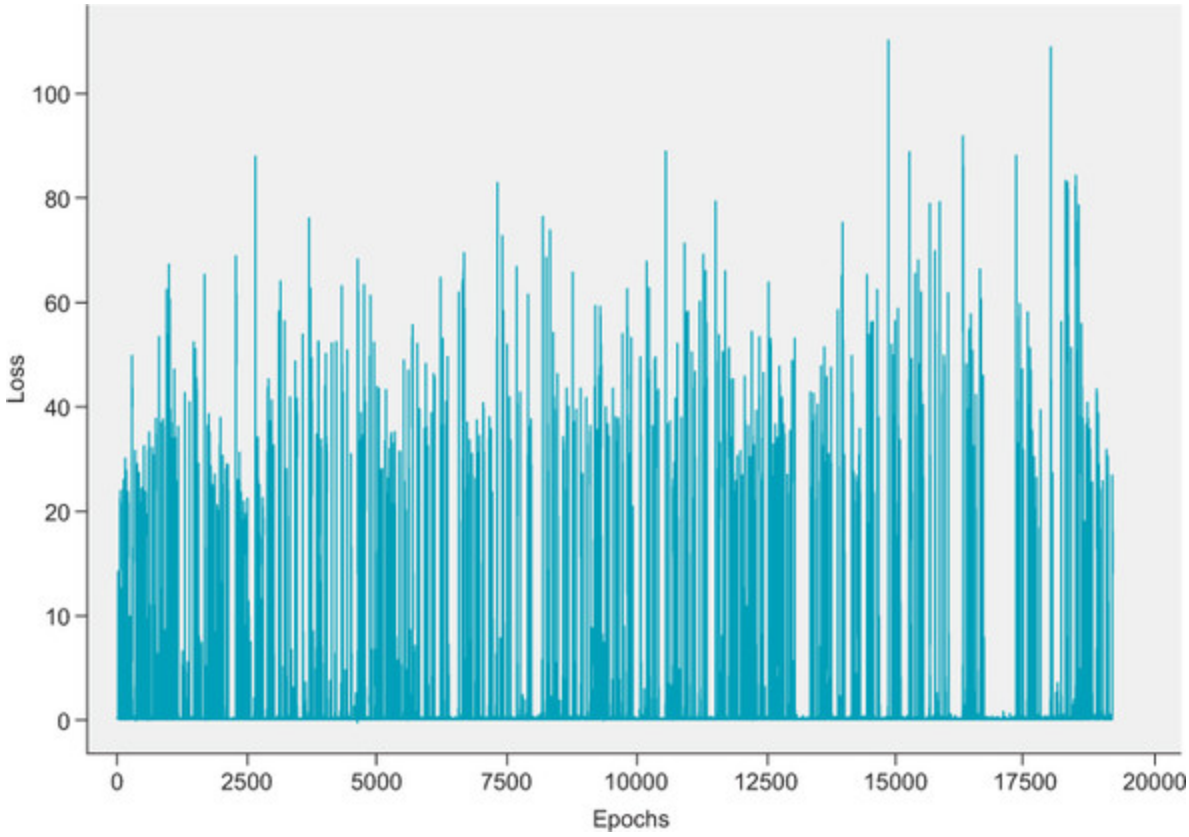
Maybe we just need to train it with the game mode set to random, and then it would really learn? Try it. Retrain it with random mode. Maybe you'll be luckier than us, but figure 3.11 shows our loss plot with random mode and 1,000 epochs. That doesn't look pretty. There's no sign that any significant learning is happening with random mode. (We won't show these results, but the model *did* seem to learn how to play with "player" mode, where only the player is randomly placed on the grid.)

**Figure 3.11. The loss plot for Q-learning in random mode, which doesn't show any signs of convergence.**




This is a big problem. Reinforcement learning won't be worth anything if all it can do is learn how to memorize or weakly learn. But this is a problem that the DeepMind team faced, and one they solved.

## 3.3. Preventing catastrophic forgetting: Experience replay